Beyond the Ratings: Unsupervised Patterns in Alone Reality TV

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Introduction to Stakeholders & Data Overview

Main question : “How do episodes differ in performance, and what underlying patterns help explain those differences?”

Stakeholders

Production team — which episode formats and locations drive viewership?

Casting producers — do survivalist profiles and gear choices shape outcomes?

Fans / community — what separates a memorable season from a forgettable one?

Data Overview

Episodes — 135 rows: IMDB ratings, vote counts, viewership per episode

Survivalists — demographics, days lasted, placement, tap-out reasons

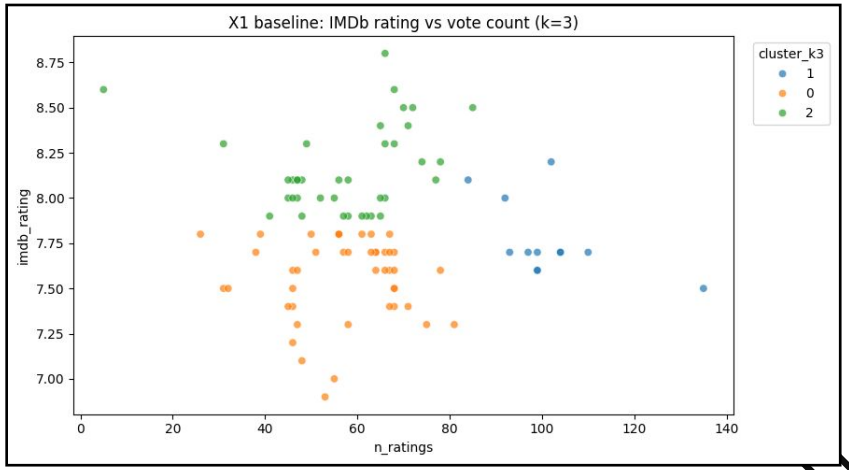
Loadouts — 10 gear items per participant across seasons

Seasons — location, country, number of survivors

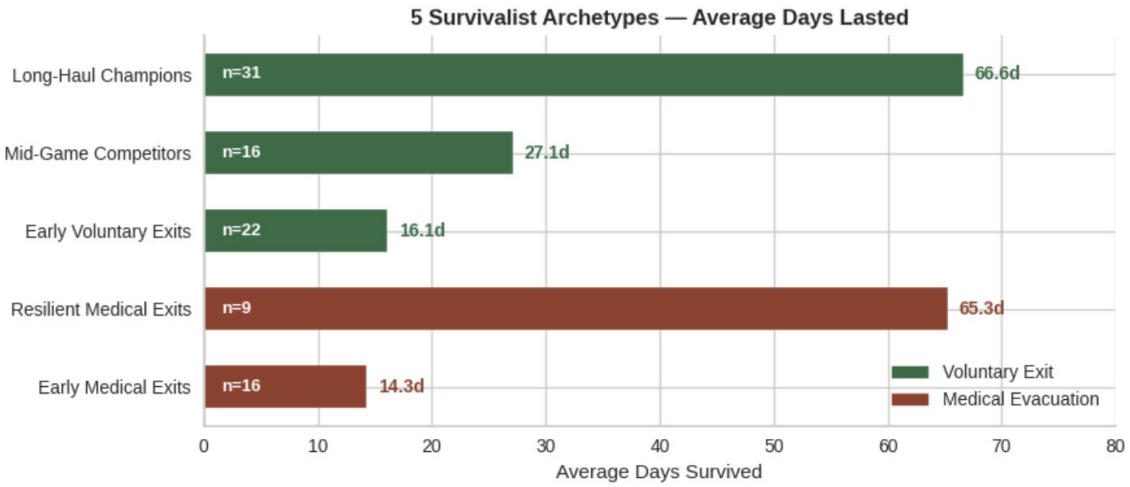
Tap-out text — exit interview quotes (reason_tapped_out)

Episode Performance Segmentation: Reach vs. Satisfaction

- We clustered IMDb ratings and vote counts to see whether episodes naturally fall into distinct performance segments that combine audience reach and viewer satisfaction.
- Clustering fits this question because it uncovers structure in episode performance without pre-labeling categories, so the data can reveal whether meaningful “types” exist.
- The output suggests episodes repeatedly form a small set of stable groups, rather than being scattered randomly across rating and engagement levels.
- A key takeaway is that reach and satisfaction do not always move together. Some episodes attract large audiences but receive moderate ratings, while others earn strong ratings from a smaller audience.



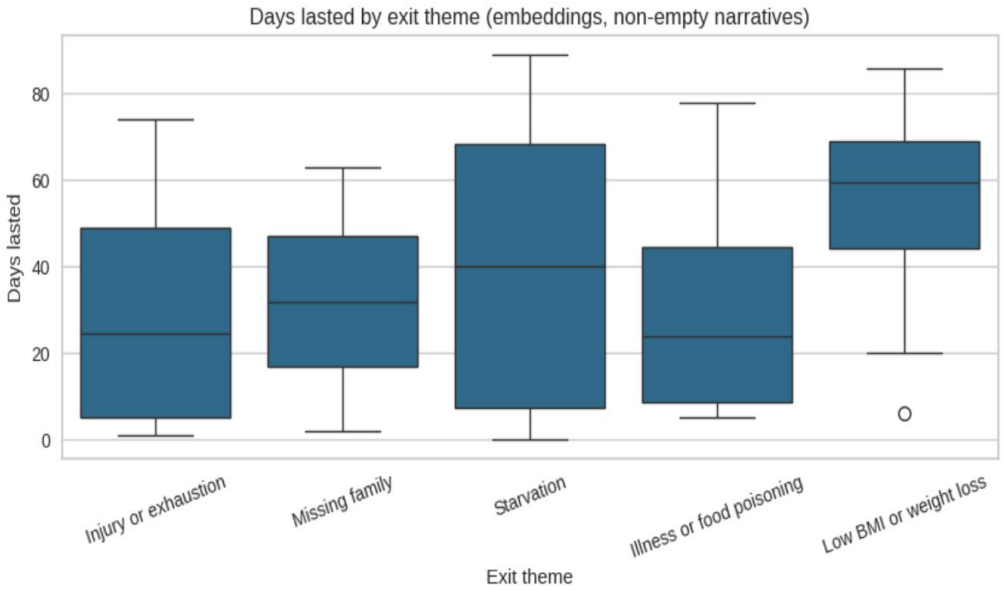
Supporting Evidence 1: Contestant Archetype



- Clustered 94 contestants on 4 features: age, days lasted, finishing placement, and medical evacuation status
- Tested k values using inertia (SSE) and silhouette → **k = 5**
- KMeans vs. Hierarchical (Ward): Hierarchical higher silhouette (0.347 vs. 0.342), used **Hierarchical labels for profiling**
- Patterns were distinct enough to assign meaningful labels (e.g., Cluster 1 mostly 1st place, 65+ days → "Long-Haul Champions")

Supporting Evidence 2: Tap-out narrative themes

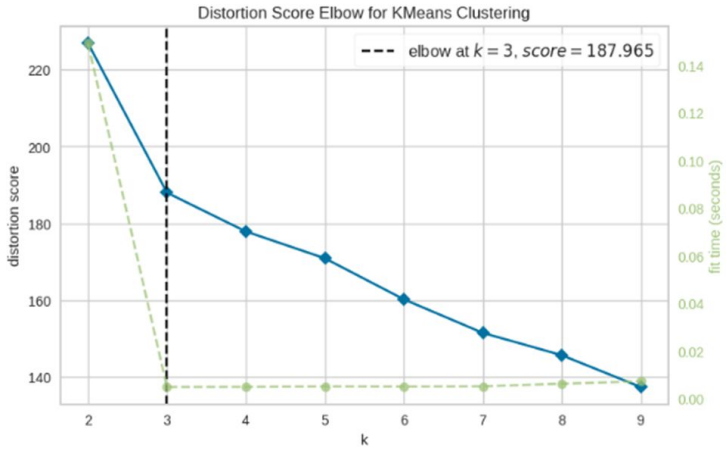
- We analyzed the tap-out narrative text which is the reason contestants give for leaving to see if exits follow repeatable patterns.
- We used text clustering so the data could form groups naturally, instead of us pre-defining exit categories ahead of time.
- The model consistently produced five interpretable exit themes: injury/exhaustion, missing family, starvation, illness/food poisoning, and low BMI/weight loss.
- To validate that these themes mattered, we compared them against days lasted.
- Key takeaway: tap-outs aren't random these themes repeat across contestants and show different survival-day profiles



- **Injury / exhaustion** — "my body is breaking down"
- **Missing family** — "I miss my kids"
- **Starvation** — "I can't get calories"
- **Illness / food poisoning** — "I can't keep food down"
- **Low BMI / weight loss** — "weight is dropping too fast"

Supporting Evidence 3: Loadout Bundles

- Each contestant picks 10 items — I built a binary matrix (1s and 0s) to compare gear choices of 94 contestants across 27 items
- Ran association rules to find which items get picked together. Core bundles like axe, ferro rod, and trapping wire showed up in 99 out of 100 bootstrap resamples
- Then clustered contestants using K-Means into three strategy archetypes: tool-heavy, shelter-first, and generalists
- Shelter-first group lasted longest at 45.5 days. Tool-Heavy lasted shortest at 31.4 days
- Contestants don't pick items randomly — they build kits. The same bundles kept showing up no matter how we changed the thresholds or resampled the data, which means these are real strategy patterns, not noise.



Elbow plot showing optimal number for k = 3

	cluster	n	top_3_items	avg_days_lasted	med_evac_%
0	0	14	Fishing gear, Saw, Pot	31.4	28.6
1	1	42	Sleeping bag, Ferro rod, Pot	45.5	26.2
2	2	38	Fishing gear, Knife, Sleeping bag	34.7	26.3

K-Means (k=3) on binary basket matrix. Validated via bootstrap (100 resamples)

Takeaways, Limitations, & Next Steps

Key takeaway: We're not claiming exit themes cause episode ratings. We're saying these are structured patterns that help interpret the show's outcomes and experience.

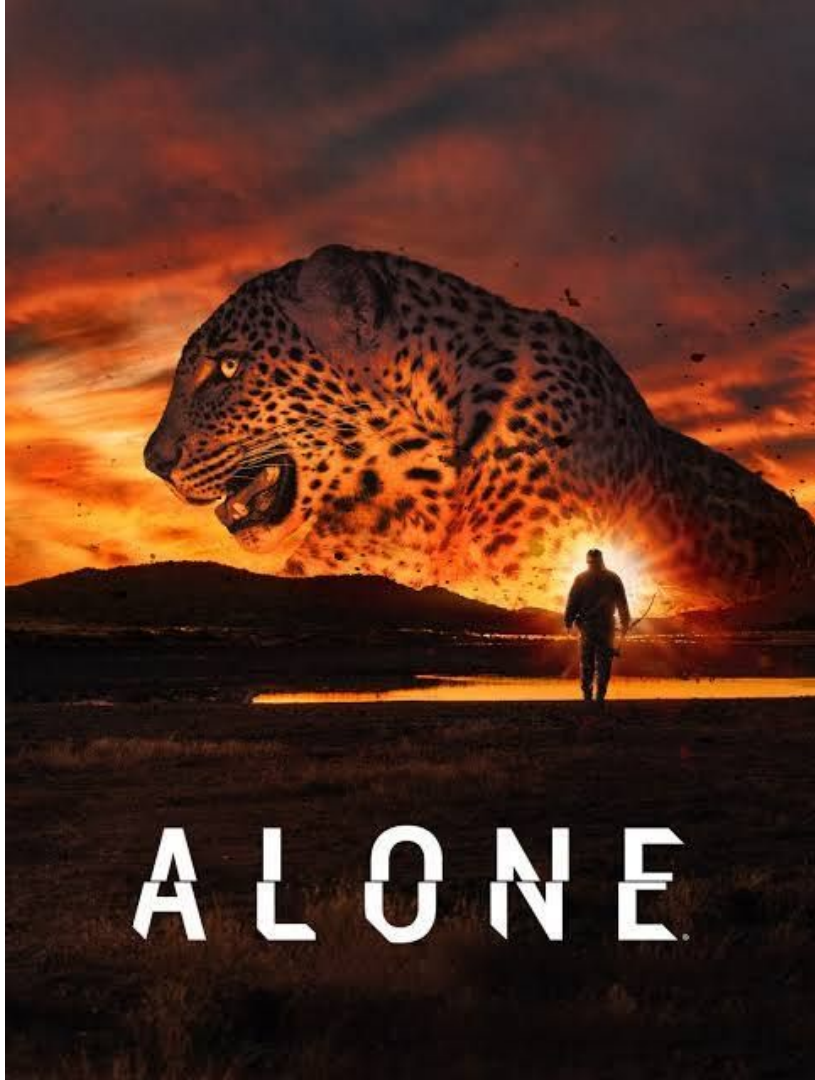
Limitations:

- Small dataset (135 episodes, ~100 survivalists)
- IMDB ratings have missing values for later seasons
- Tap-out text is short and inconsistent — limits NLP depth
- No causal claims, only pattern identification

Next steps:

- Incorporate social media/Reddit sentiment as richer text data
- Supervised model using clusters as features to predict viewership
- Deeper loadout-to-outcome analysis with more seasons

Thanks

Any questions?

https://github.com/MikaIsmayilov/B02_Alone_TVShow_Unsupervised_Project/tree/main